

13. (g) Implementation Plan: B.Tech. in Biomedical Engineering with Specialization in Machine Intelligence

Semester – I					
Code	Course Title	Hours/ Week			C
		L	T	P	
21LEH101T	Communicative English	2	1	0	3
21MAB101T	Calculus and Linear Algebra	3	1	0	4
21PYB101J	Physics: Electromagnetic Theory, Quantum Mechanics, Waves and Optics	3	1	2	5
21MES102L	Engineering Graphics and Design	0	0	4	2
21EES101T	Electrical and Electronics Engineering	3	1	0	4
21BMC101J	Biomedical Sensors	2	0	2	3
21CYM101T	Environmental Science	1	0	0	0
21PDM101L	Professional Skills and Practices	0	0	2	0
21LEM101T	Constitution of India	1	0	0	0
Total Credits					21

Semester – II					
Code	Course Title	Hours/ Week			C
		L	T	P	
21LEH102T	Chinese	2	1	0	3
21LEH103T	French				
21LEH104T	German				
21LEH105T	Japanese				
21LEH106T	Korean				
21LEH107T	Spanish				
21GNH101J	Philosophy of Engineering				
21MAB102T	Advanced Calculus and Complex Analysis	3	1	0	4
21CYB101J	Chemistry	3	1	2	5
21CSS101J	Programming for Problem Solving	3	0	2	4
21BTB104T	Biology-Human Physiology and Anatomy	2	0	0	2
21MES101L	Basic Civil and Mechanical Workshop	0	0	4	2
21PDM102L	General Aptitude	0	0	2	0
21GNM101L	Physical and Mental Health using Yoga	0	0	2	0
21GNM102L	NSS				
21GNM103L	NCC				
21GNM104L	NSO				
Total Credits					22

Semester – III					
Code	Course Title	Hours/ Week			C
		L	T	P	
21MAB201T	Transforms and Boundary Value Problems	3	1	0	4
21BMC202T	Biomedical Signals and Systems	3	0	0	3
21BMC203J	Electric and Electronic Circuits	3	0	2	4
21BMC204J	Digital Logic for Medical Systems	2	0	2	3
21PDH201T	Social Engineering	2	0	0	2
21PYS202T	Medical Physics	3	0	0	3
21LEM201T	Professional Ethics	1	0	0	0
21PDM201L	Verbal Reasoning	0	0	2	0
21LEM202T	Universal Human Values– Understanding Harmony and Ethical Human Conduct	2	1	0	3
Total Credits					22

Semester – IV					
Code	Course Title	Hours/ Week			C
		L	T	P	
21MAB202T	Numerical Methods	3	1	0	4
21CSE206T	Artificial Intelligence	2	1	0	3
21BMC205J	Integrated Circuit Design for Bioinstrumentation	2	0	2	3
21BMC206J	Biomedical Instrumentation	3	0	2	4
21BMC207J	Biomaterials and Tissue Interaction	2	0	2	3
E	Professional Elective – I				3
21DCS201P	Design Thinking and Methodology	1	0	4	3
21PDM202L	Critical and Creative Thinking Skills	0	0	2	0
Total Credits					23

Semester – V					
Code	Course Title	Hours/ Week			C
		L	T	P	
21MAB301T	Probability and Statistics	3	1	0	4
21BMC301J	Biomedical Signal Processing	3	0	2	4
21BMC302J	Microcontrollers and its Application in Medicine	3	0	2	4
21BMC303T	Principles of Medical Imaging	3	0	0	3
E	Professional Elective – II				3
O	Open Elective – I				3
21PDM301L	Analytical and Logical Thinking Skills	0	0	2	0
21LEM301T	Indian Art Form	1	0	0	0
21GNP301L	Community Connect	0	0	2	1
Total Credits					22

Semester – VI					
Code	Course Title	Hours/ Week			C
		L	T	P	
21CSS303T	Data Science	1	1	0	2
21BMC304J	Medical Image Processing	2	0	2	3
21BMC305T	Biocontrol Systems	3	0	0	3
E	Professional Elective – III				3
E	Professional Elective – IV				3
21BMP303T	MOOC	3	0	0	3
21BMP302L	Project	0	0	6	
O	Open Elective – II				3
21PDM302L	Employability Skills and Practices	0	0	2	0
21LEM302T	Indian Traditional Knowledge	1	0	0	0
Total Credits					20

Semester – VII					
Code	Course Title	Hours/ Week			C
		L	T	P	
21GNH401T	Behavioral Psychology	2	0	2	3
E	Professional Elective – V				3
E	Professional Elective – VI				3
21BMC401J	Biomechanics	2	0	2	3
21BMC402J	Biomedical Equipments for Clinical Applications	2	0	2	3
O	Open Elective – III				3
Total Credits					18

Semester – VIII					
Code	Course Title	Hours/ Week			C
		L	T	P	
21BMP401L	Major Project	0	0	30	15
21BMP402L	Major Project	0	0	20	
21BMP403L	Internship#	0	0	10	
Total Credits					15

#Students have to register either 21BMP401L or 21BMP402L and 21BMP403L both in eighth semester